Security

Session 8



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Objectives

- *Describe security threats*
- *Explain security in Apache Web server*
- *Explain the process of authenticating a user and a host*
- *Explain the use of a proxy*
- *Explain the use of a firewall*

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Security Threats

- Insecure Architectures
- Broadcast Networks
- Centralized Servers
- Unused Services and Open Ports
- Inattentive Administration
- Inherently Insecure Services
- Bad Passwords
- Vulnerable Client Applications

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Authentication Ways

- Apache server authenticates a HTTP request in three ways:
  1. Access Control
     - Restricts access to incoming HTTP request
  2. Authentication
     - Process of verifying the correctness of the username and the password
  3. Authorization
     - Allows access to a resource protected by authentication

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Authentication Directives

- The `core` module supports the authentication directives
- Some of the directives that enable authentication are as follows:
  - `AuthName` Directive
  - `AuthType` Directive
  - `Require` Directive
  - `AuthUserFile` Directive
  - `AuthGroupFile` Directive

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AuthName Directive

- Specifies the name of authorization realm for a directory

Syntax

```
AuthName auth-domain
```

where,

`AuthName` - specifies the name of the authentication information

`auth-domain` - specifies the argument for the directive

Snippet

```
AuthName "Security"
```

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AuthType Directive

- Authentication methods that are available
  - `Basic` – Requires authentication and authorization with a username and a password
  - `Digest` - requires a password file to verify the username and the password

Syntax

```
AuthType Basic|Digest
```

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Require Directive

- Specifies the authorized users who are allowed access to a resource

Syntax

```
Require entity-name [entity name]
```

where,

`Require` - specifies access permissions

`entity-name` - specifies the users who have access to the resource

Syntax	Permissions
`Require user userid [userid]...`	Allows the specified users to access the resource
`Require group group-name [group-name] ...`	Allows only the users mentioned in the named groups to access the resource
`Require valid-user`	All valid users can access the resource

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AuthUserFile Directive

- Defines name of the text file that contain username and password for user authentication
- This file stores a list of usernames and corresponding passwords

Syntax

```
AuthUserFile file-path
```

where,

`file-path` - specifies the path of the user file

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AuthGroupFile Directive

- Defines the name of the text file that stores a list of user groups
- Every line in the text file contains a group name followed by a colon and the usernames separated by spaces

Syntax

```
AuthGroupFile file-path
```

where,

`file-path` - defines the path of the text file containing the user groups

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Basic Authentication

- If the client browser supports basic authentication, it prompts the user for a username and a password
- If the user enters
  - Username = John
  - Password = 123

Browser generates a new request including the Authorization header.

i.e.

```
t90jhismlvghdftHLkpw=
```

```
Authorization: BASIC t90jhismlvghdftHLkpw=
```

Snippet

```
<Location /myfiles>
  AuthName "Protected Documents"
  AuthType Basic
  AuthUserFile /usr/local/apache2/passwords/passwdfile
  AuthGroupFile /usr/local/apache2/passwords/grpfile
  Require valid-user
</Location>
```

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Digest Authentication

- Improves the security of passwords
- Uses the module `mod_auth_digest`
- Transmits a digest of the password instead of the password

Directive	Function
`AuthDigestProvider`	Specifies the authentication provider for the location.
`AuthDigestAlgorithm`	Specifies the algorithm to use for calculating the challenge and response hash. Challenge hash is used to encrypt the authentication while response hash is used to verify or decrypt the authentication.
`AuthDigestShmemSize`	Defines the memory space to be allocated at startup. Apache server uses this memory space to track client activities.

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Creating and Maintaining password files

- Password file is required to restrict access to a file using a password

Syntax

```
htpasswd option passwdfile username
```

where,

`option` - specifies a list of available options

`passwdfile` - specifies the name of the password file that stores the username and password

`username` - specifies the username to be created or updated

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Creating and Maintaining password files

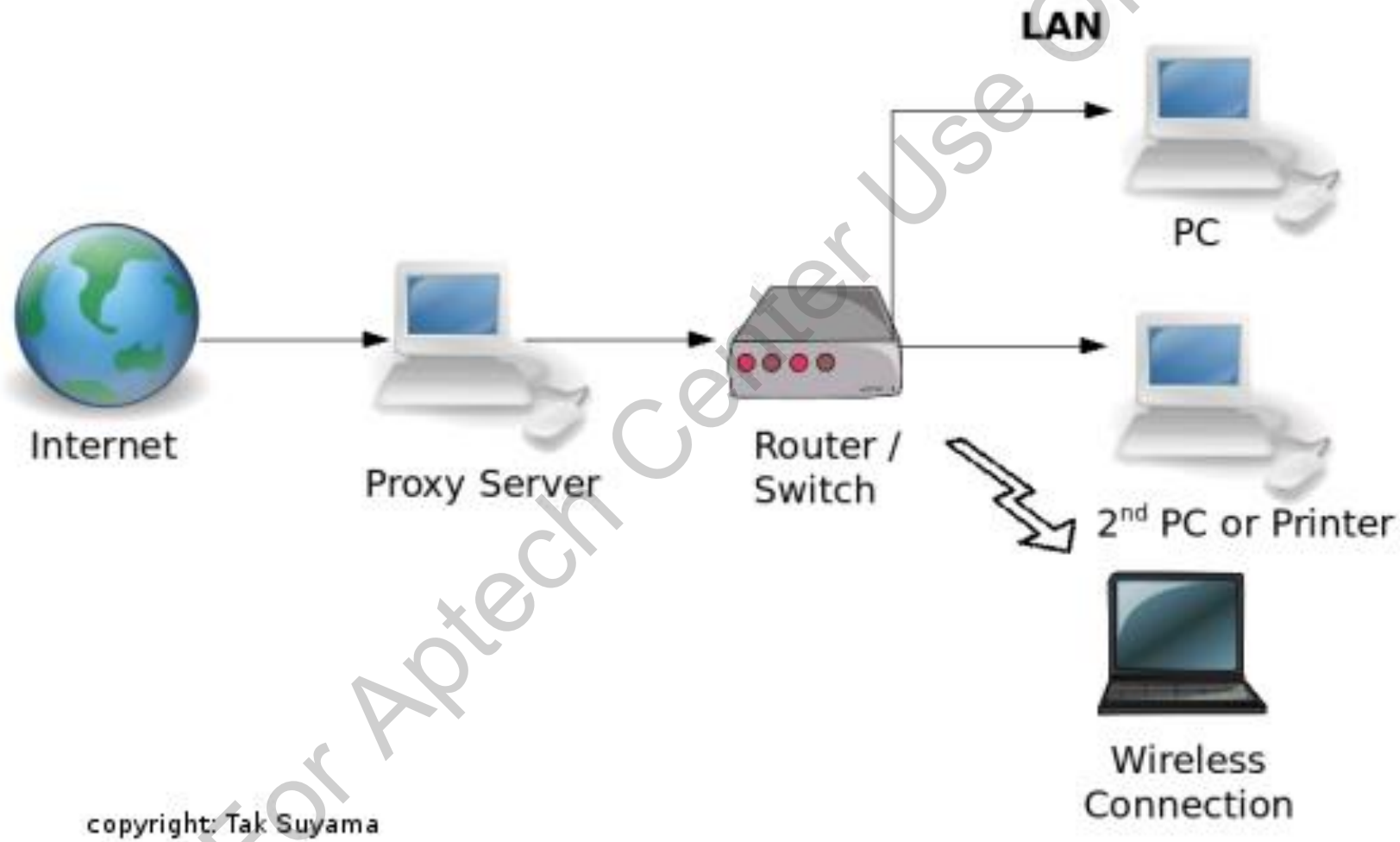
Snippet

```
/usr/local/apache2/bin/htpasswd -c /usr/local/apache2/passwords john
```



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Proxy Server



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Proxy Server

- Acts as an intermediate server between a client browser and the HTTP server
- Main objective is to filter client requests
- Intercepts all the client requests and fulfills them if the resource requested by the client exists on the proxy
- If unable to fulfill the client request, it forwards the request to the remote HTTP server

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Four basic modules of Proxy

- mod_proxy
  - Specifies the proxy or the gateway for Apache server
- mod_proxy_connect
  - Supports the CONNECT HTTP method. Apache server uses this module to pass SSL request through proxy servers
- mod_proxy_ftp
  - Supports to proxy FTP requests using the GET method only
- mod_proxy_http
  - Enables proxy for HTTP and HTTPS requests

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Proxy Directives

- `<Proxy>` Directive
- `ProxyRequests` Directive
- `ProxyIOBufferSize` Directive
- `ProxyDomain` Directive
- `ProxyBlock` Directive
- `ProxyMaxForwards` Directive

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Types of Proxies

1. Forward Proxy

- Receives the request from a client and forwards it to the remote server
- Reduces load on the network between the proxy and the remote server
- Used for
  - Security
  - Caching
  - Filtering

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2. Reverse Proxy

- Enables controlled access from the Internet to the servers behind a firewall
- Browsers interpret a reverse proxy like a regular Web server – so no additional configuration required
- Used for
  - Internet access to a server behind firewall
  - Load balancing among backend servers
  - Caches for backend server

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To configure Apache as Proxy Server

Snippet

```
Listen 8080
User httpd
Group httpd
ProxyRequests On
```

- To secure proxy server

Snippet

```
<Proxy 192.127.12.10>
  Order Deny, Allow
  Deny from all
  Allow from 127.0.0.1
</Proxy>
```

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Firewall

- Provide protection to networks against outside attackers by protecting the network
- Exist between the computer and the network
- Allow only relevant and necessary data
- Two types
  - Network-Level Firewalls: Packet Filters
    - Provides routers with packet filtering capability
  - Application-Proxy Firewalls/Application Gateways
    - Establishes connection between outside clients and internal network

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Configuring Firewall

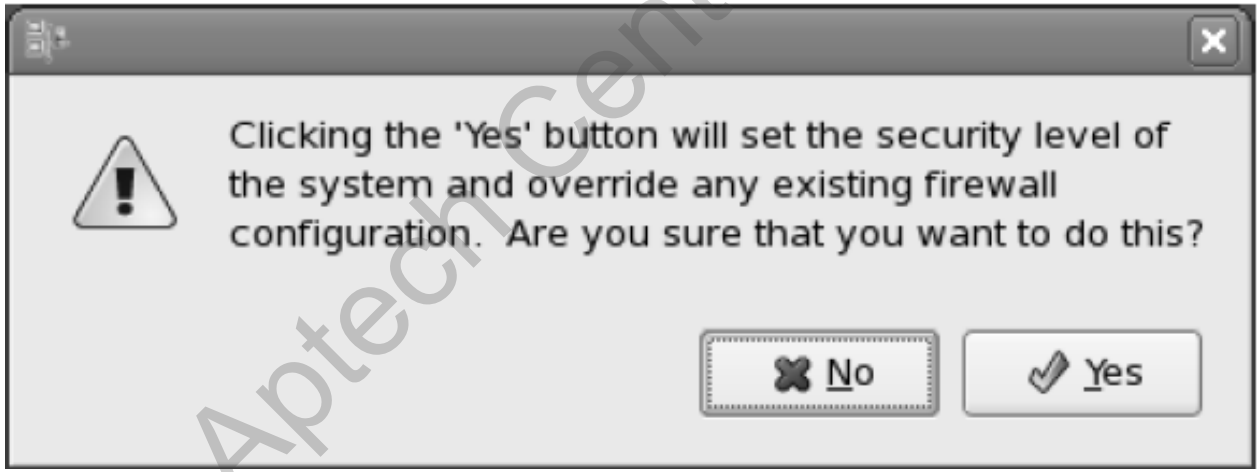
1. Select System → Administration → Security Level and Firewall. The Security Level Configuration dialog box appears



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Configuring Firewall

2. In the Trusted services pane, check the FTP, Mail (SMTP), SSH, Secure WWW (HTTPS), Telnet, and WWW (HTTP) check boxes
3. Click Apply. Red Hat Enterprise Linux prompts for confirmation



4. Click Yes
5. Click OK

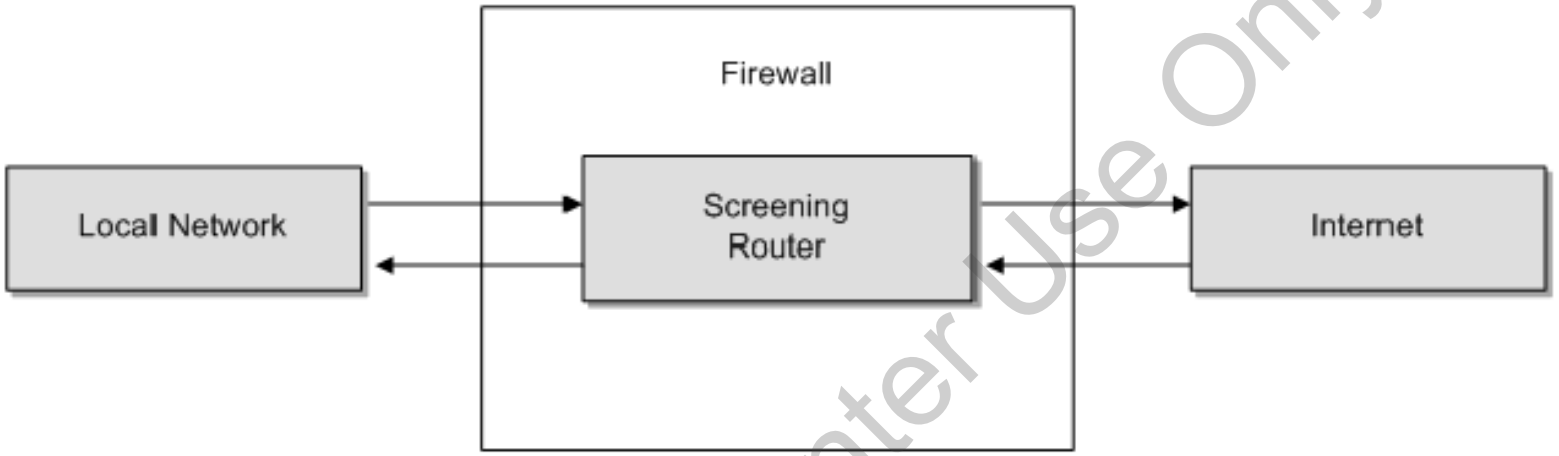
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Firewall Architecture Models

- Screening Router Architecture
- Dual-Homed Host Architecture
- Screened Host Architecture
- Screened Subnet Architecture

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Screening Model Architecture



- Consists of
  - Local network of computers
  - A screening router
  - An external network, such as the Internet
- The screening router functions:
  - Firewall
  - Allows data packets acceptable by the network to pass through it

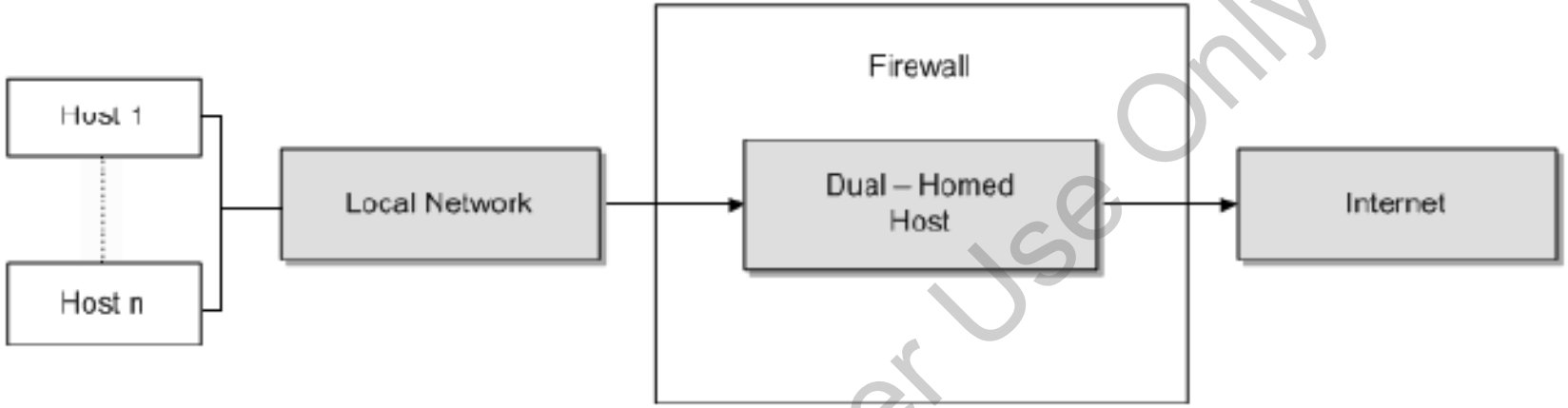
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Screening Model Architecture

- Advantages
  - Transparent
  - Easier to use and less expensive
- Disadvantages
  - Cannot handle all types of traffic
  - Configuration of this architecture is difficult
  - Configuration of this architecture is difficult
  - Does not contain user authentication

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Dual-Homed Host Architecture



- Local network and Internet communication is permitted in two ways:
  - Account on the Dual-Homed Host machine
    - Enables the user to logon to the Dual-Homed Host machine to access the Internet
  - Proxy Programs
    - Executes on the Dual-Homed Host machine to enable access to the Internet

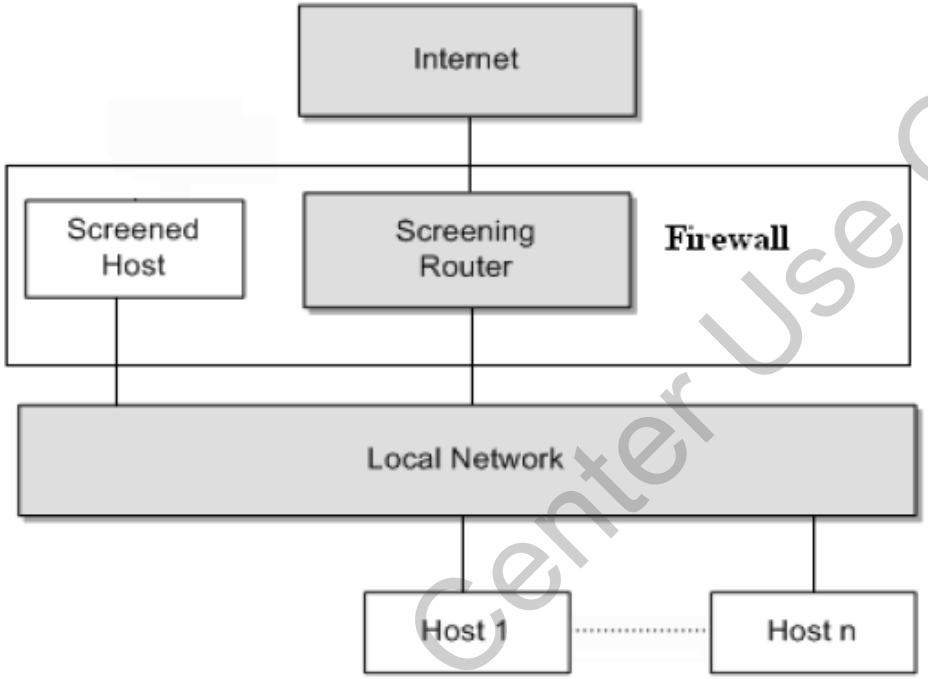
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Dual-Homed Host Architecture

- Advantages
  - More secure
  - Internal network structure is not accessible
- Disadvantages
  - Not user friendly
  - Needs multiple proxies
  - Involves change in user behavior

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Screened Host Architecture



- Consists of the screening router and the screened host
- Packet filtering by the screening router
- Services given by the screened host

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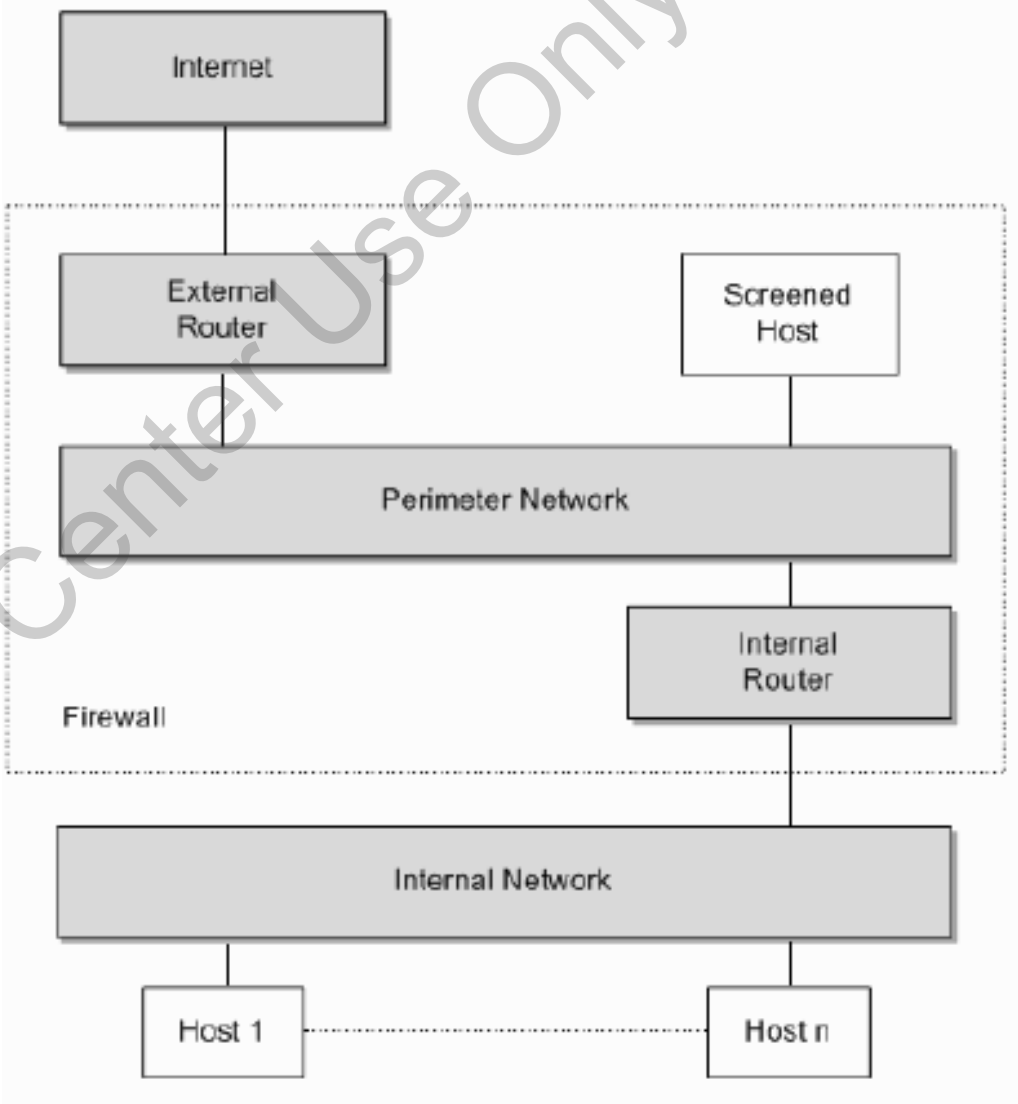
Screened Host Architecture

- Advantages
  - Multiple authentication
  - Transparent outbound access
  - Restricted inbound access
- Disadvantages
  - Internal structure not hidden
  - Single point of failure (router)
  - No protection from compromised gateway

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Screened Subnet Architecture

- Consists of a screened host connected to a network called as perimeter network
- Perimeter net offers an additional layer of security for your internal systems
- Router protects the perimeter net and the internal network from unauthorized Internet users
- Provides a better security



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Screened Subnet Architecture

- Advantages
  - Adaptable form
  - Internal network structure is not accessible
  - Better security
- Disadvantages
  - All security functions provided by gateway, a single point of security failure

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Summary

- User authentication requires username and password for permitting access to a resource
- Apache supports `basic` and `digest` authentication methods
- Password files are used to verify and authenticate the username and the password
- Proxies are intermediate servers connected between a client and a remote server
- Firewalls can be configured to block data from specific locations and allow only relevant and necessary data

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